Flow Matching for Virtual HER2 Staining

Staining Flow Swimmers

NeuroML Final Project Report

Abstract

We study H&E-conditioned generation of HER2 immunohistochemistry (IHC) with Flow Matching. Experiments on MIST-HER2 showed weak use of the H&E condition, sensitivity to pair registration, and systematic under-reconstruction of strong brown DAB staining. Variance-preserving paths and batch optimal-transport coupling did not improve all objectives simultaneously. A smaller 256-pixel crop increased paired similarity, whereas OT-CFM improved distributional metrics. On the better-aligned HER2match dataset, direct H&E-to-IHC transport improved reconstruction, but high SSIM could coexist with false or missing marker signal. Stain-aware sampling and a staining-sensitive loss reduced false staining without solving missing strong expression. The central result is that alignment and marker-aware evaluation are as important as the probability path.

1. Task and experimental design

The task is to generate a virtual HER2 IHC image from an H&E image while preserving tissue morphology and reproducing the amount and spatial localization of HER2 staining. Virtual staining is treated as a research problem, not a diagnostic replacement. Conditional Flow Matching (CFM) regresses a time-dependent velocity field and integrates it with an ODE solver at inference [1]. We tested noise-to-IHC generation with H&E conditioning, direct H&E-to-IHC transport, alternative probability paths, minibatch OT coupling, crop scale, foundation-model features, and data/loss interventions.

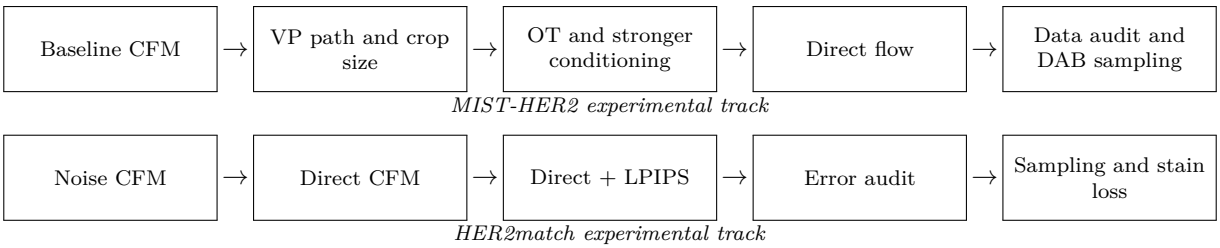


Figure 1: Experimental pipelines for MIST-HER2 and HER2match.

Data and evaluation

MIST-HER2 was split by tissue group into 3,805 training, 784 validation, and 1,053 test pairs. The dataset contains paired H&E-IHC patches for four breast-cancer biomarkers; however, because the stains were acquired from adjacent tissue sections, local morphological inconsistencies may remain even after registration [3]. Images were cropped or resized to a fixed spatial resolution and normalized to the model input range. Our main experiments used 512×512 crops, while a 256×256 crop ablation tested whether a smaller field of view simplified local reconstruction. To prevent data leakage, the split was performed by tissue group rather than by individual patch. This resulted in 40 training, 9 validation, and 12 test groups with no overlap between the splits.

HER2match contains 21,172 same-tissue H&E/HER2 tiles, divided into 11,610 training, 3,582 validation, and 5,980 test pairs [5, 4]. Compared with MIST-HER2, its more accurately aligned pairs provide a cleaner setting for separating model limitations from registration-related errors.

Evaluation combined complementary measures of reconstruction and distributional quality. SSIM, PSNR, and MAE quantify paired image fidelity; FID and KID measure similarity between generated and real image distributions; and LPIPS was reported where available as a perceptual-distance metric. We additionally measured the tissue-masked DAB-positive area to detect false or missing brown staining that may not be reflected by conventional image-quality metrics. DAB area is used only as an optical proxy for staining intensity and must not be interpreted as a clinical HER2 score.

2. MIST-HER2 experiments

Baseline and model variants

The baseline was a conditional noise-to-IHC Flow Matching model. A U-Net predicted the velocity between Gaussian noise and real HER2 IHC, while the paired H&E image was provided as a spatial condition. Although the generated images had plausible IHC-like colour and texture, different H&E inputs frequently produced similar pale predictions. The model therefore captured the target appearance better than the input-specific morphology and staining.

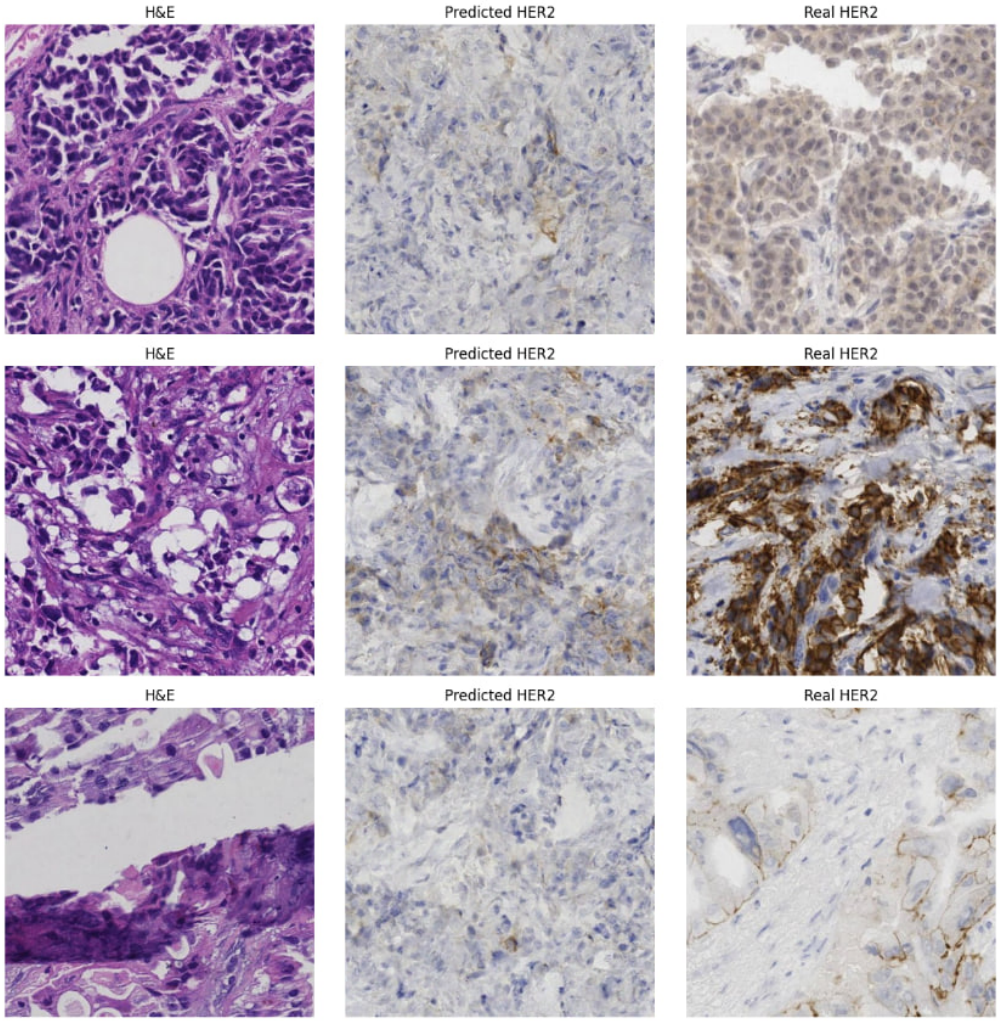


Figure 2: Baseline conditional CFM on MIST-HER2. Predictions for different H&E inputs converge toward similar pale IHC appearances and underestimate strong HER2 staining, indicating weak spatial conditioning and target averaging.

We tested the following modifications:

- **Stronger spatial conditioning.** Additional multi-scale conditioning paths exposed the U-Net more directly to H&E morphology. Input dependence remained limited, suggesting that conditioning capacity alone was not the main bottleneck.
- **Exact OT-CFM.** Random noise-target pairing within each minibatch was replaced by an exact optimal-transport assignment implemented through TorchCFM [2, 6]. The paired H&E condition was reordered together with its IHC target. OT improved distributional metrics relative to the controlled VP-CFM and crop-256 ablations, but did not remove weak H&E dependence.
- **Phikon conditioning.** Features from the pretrained Phikon histopathology model were added as high-level morphological context. Selected examples showed stronger structural correspondence, but staining fidelity did not improve consistently. The saved metrics cover only a small selected subset and are therefore excluded from the full-test table.
- **Structure-sensitive losses.** Feature and structural losses were added to penalize morphological disagreement beyond velocity matching. They could improve SSIM while producing smoother or mask-like outputs, showing that structural similarity does not guarantee correct HER2 staining.

- **Direct H&E-to-IHC transport.** Gaussian noise was replaced by H&E as the source distribution. Direct transport increased dependence on the input structure, but predictions often became mask-like and were particularly unreliable for locally misregistered pairs.

These experiments shifted the main hypothesis from insufficient conditioning alone to two data limitations: imperfect local H&E-IHC alignment and an imbalanced target-stain distribution.

Path and crop ablations

Two controlled ablations retained the conditional U-Net and changed one component of the baseline.

Variance-Preserving CFM replaced linear interpolation with a trigonometric variance-preserving path. This tested whether the baseline failure resulted from unsuitable path geometry. VP-CFM degraded both paired and distributional metrics.

Crop-256 reduced the training crop from 512×512 to 256×256. Smaller crops reduced memory use and simplified local correspondence but contained less global context. Crop-256 improved SSIM but continued to underestimate strong DAB staining.

A conditional CFM trained on HER2match was also evaluated as a cross-dataset control. Because its training data and preprocessing differed from MIST, it measures domain transfer rather than a controlled MIST ablation.

Method	n	**SSIM** ↑	**FID** ↓	**KID**$\times 10^3$ ↓
Earlier full-test experiments				
CFM baseline, crop 512	1,053	0.1520	163.0	95.0
CFM trained on HER2match	1,053	0.1438	217.0	211.0
Controlled path and crop ablations				
CFM baseline, crop 256	1,053	**0.1700**	222.61	255.18
Variance-Preserving CFM, crop 512	1,053	0.1447	367.41	469.09
Exact OT-CFM, crop 512	1,053	0.1682	**186.58**	**191.55**

Table 1: Our full-test MIST-HER2 results. The two blocks were produced by different saved evaluation pipelines and are not used for strict cross-block ranking. Within the controlled ablations, crop-256 achieved the highest SSIM, while Exact OT-CFM achieved the best FID and KID.

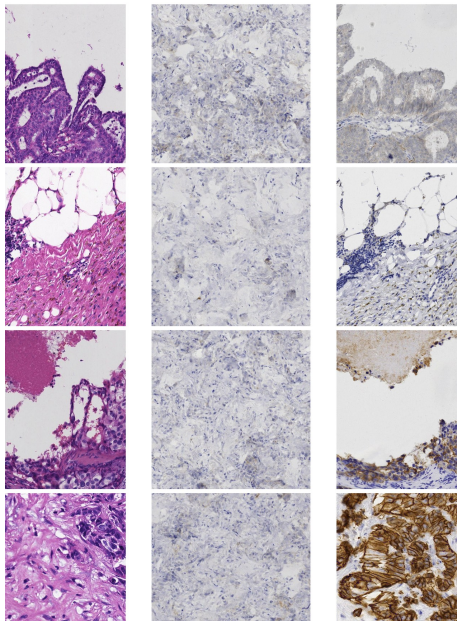


Figure 3: Crop-256 predictions across increasing real DAB levels. Weakly stained cases can appear plausible, while strong target staining is systematically underestimated.

Interpretation

The experiments separated three objectives. Crop-256 improved paired similarity, Exact OT-CFM improved distributional metrics within the controlled ablations, and direct or feature-guided variants increased visible H&E dependence in selected cases. No method simultaneously preserved input morphology, matched the target distribution, and reproduced HER2 staining.

Two explanations remained. First, local misregistration can penalize a structurally reasonable prediction when the paired IHC target is displaced. Second, the predominance of weakly stained training patches can favour pale average outputs. These hypotheses motivated the alignment and DAB audit described in the next section.

3. MIST audit and data interventions

A dataset audit separated tissue coverage, pair alignment, and DAB intensity. All 1,053 test images contained adequate tissue, so empty patches were not the main bottleneck. Alignment quality varied substantially; a conservative clean diagnostic subset contained 263 pairs, but the full test split remained the primary benchmark. Errors increased on poorly aligned pairs, and high-DAB regions were frequently under-stained.

Interventions

- DAB-balanced sampling targeted weak, moderate and strong staining while retaining negative examples.
- DAB plus alignment balancing additionally down-weighted poorly aligned training pairs.
- Both runs used the same crop-512 conditional CFM backbone and were evaluated on the same DAB-stratified subset.

Variant	n	SSIM	PSNR	MAE	FID	KID$\times 10^3$	DAB error
DAB-balanced	350	0.1694	13.70	0.1728	365.75	454.73	0.1315
DAB + alignment	350	**0.1757**	**13.73**	**0.1701**	**327.90**	**382.83**	0.1314

Table 2: Diagnostic evaluation on 350 stratified test images: 100 each from $\leq 1\%$, 1–10%, and 10–30% real DAB, plus 50 above 30%. These values are not directly comparable with Table 1.

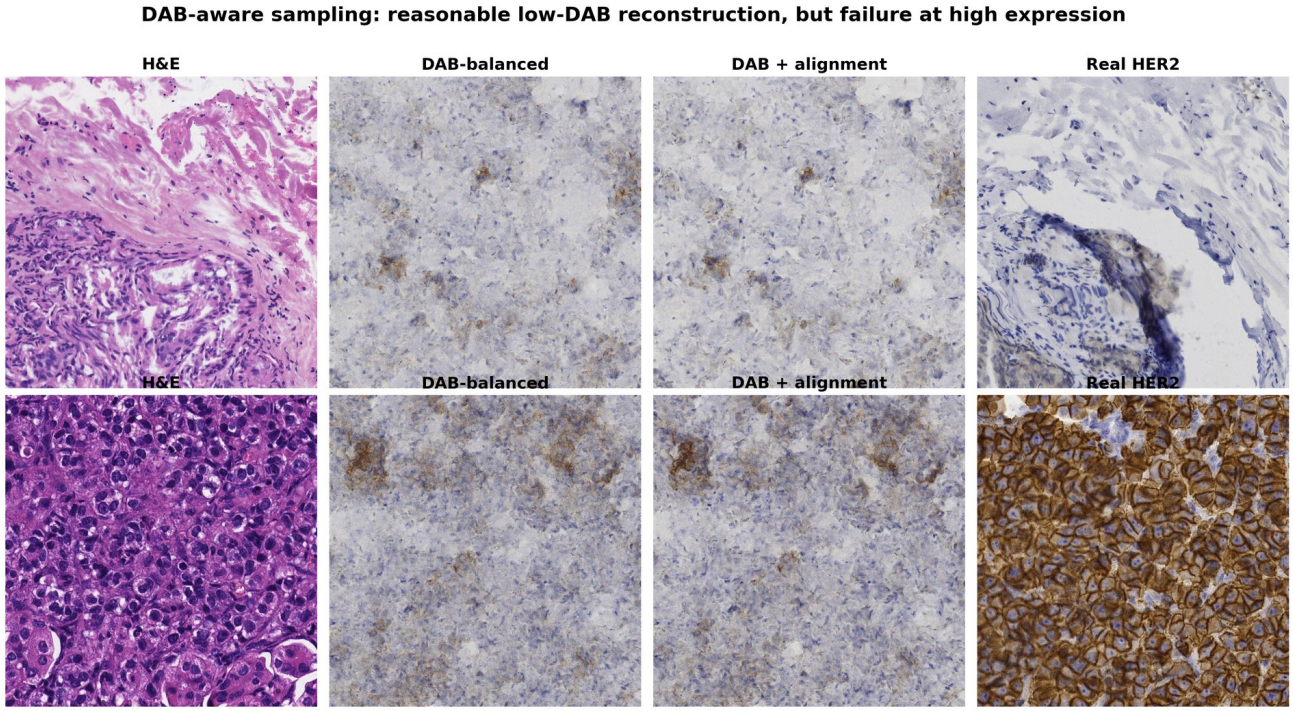


Figure 4: Representative low- and high-DAB cases. Balancing retains reasonable low-DAB appearance but does not recover extensive real staining.

Alignment-aware sampling produced a small improvement in SSIM, FID and KID, but almost no change in mean DAB-fraction error. Better example selection helps global image statistics, yet the model still collapses strong expression toward pale average outputs.

4. HER2match experiments

HER2match was used to test whether better registration changes the model ranking. Noise CFM was followed by direct H&E-to-IHC CFM and a Direct + LPIPS ablation. Direct transport improved reconstruction, motivating a

targeted audit. The audit identified false brown staining in weakly stained tissue, missing positive signal, and loss of fine contrast. Two controlled fixes retained the direct architecture: stain-aware crop sampling and a DAB-sensitive endpoint loss.

Model	SSIM ↑	LPIPS ↓	Missing stain ↓	False stain ↓
Noise CFM	0.319	0.502	7.65%	0.46%
Direct CFM	0.397	0.456	7.61%	0.90%
Direct + LPIPS	0.388	0.482	7.93%	**0.01%**
Direct + sampling	**0.408**	0.461	7.62%	0.19%
Direct + stain loss	0.390	**0.444**	7.80%	0.58%

Table 3: HER2match comparison on the same 252 aligned pairs, with 63 pairs per slide-level HER2 score. Stain areas are optical measurements within tissue.

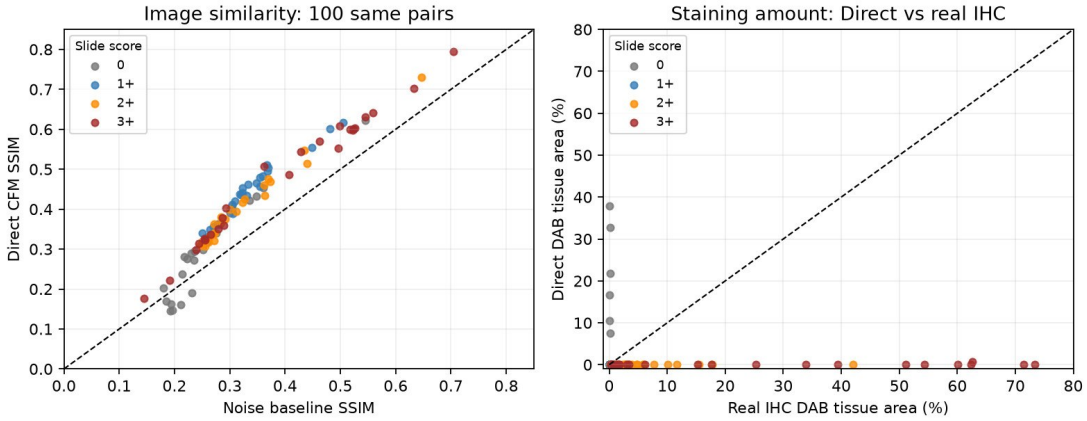


Figure 5: Direct CFM improves SSIM for most audited pairs (left), yet predicted DAB area remains near zero for many strongly stained targets (right).

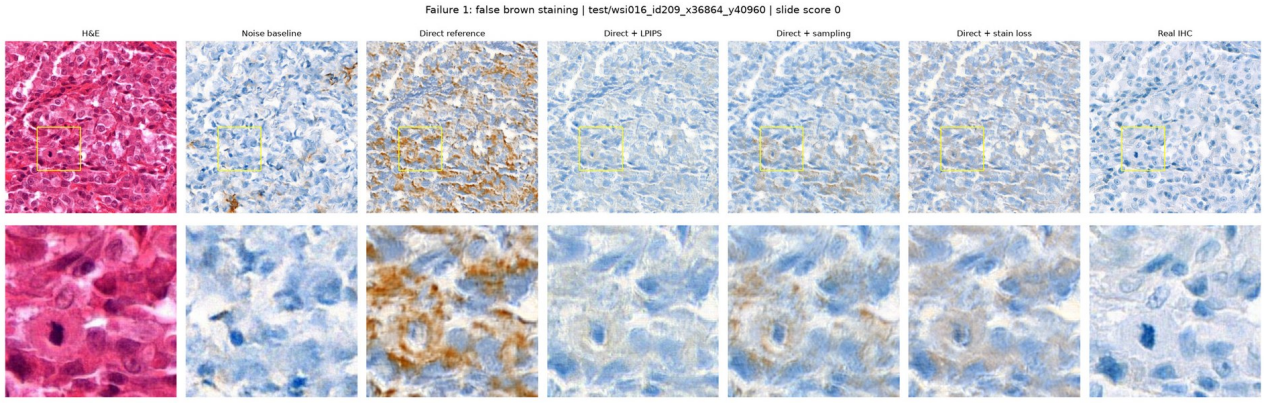


Figure 6: A controlled improvement: false staining falls from 37.89% with Direct CFM to 6.01% with sampling and 5.09% with stain loss; real IHC is 0.08%.

Sampling achieved the best SSIM and reduced mean false-stain area from 0.90% to 0.19%; stain loss achieved the best LPIPS. Neither reduced average missing-stain area. In the strongest selected positive case, real IHC contained 73.35% stained tissue, while sampling predicted 0.017% and stain loss 0.55%. Thus, structural reconstruction and marker fidelity must be reported separately.

5. Conclusions, limitations and next steps

Main findings

- Alignment is a central experimental variable. Better-aligned HER2match pairs supported stronger paired reconstruction than MIST, but did not guarantee correct staining.
- No Flow Matching variant dominated every metric. Exact OT-CFM improved MIST FID/KID, whereas crop-256 and direct transport favored paired similarity.

- DAB-aware interventions reduced false positives but did not restore strong positive HER2 signal. High SSIM can coexist with biologically important failure.

Limitations

Cross-dataset comparisons do not isolate alignment from cohort, preprocessing and split differences. MIST ablation metrics were produced under more than one resizing/evaluation protocol; only internally consistent values were compared above. The DAB study used 350 stratified test images and checkpoints stopped at 7,500 updates, so it is diagnostic rather than a full benchmark. HER2 slide scores are not patch-level clinical labels, DAB thresholds are heuristic, and selected failure cases do not estimate population frequency. No pathologist-validated membrane-localization metric was available.

Next steps

- Use one fixed split, checkpoint rule, solver and metric implementation for every ablation.
- Stratify evaluation by real DAB intensity and report false and missing staining separately from SSIM and LPIPS.
- Improve registration or use alignment-robust patch/feature losses; test one intervention at a time.
- Add pathologist-reviewed membrane annotations or a validated HER2-localization metric before clinical interpretation.

Team contributions

Member	Primary contribution
Vasilii Gorelov	MIST baseline and architecture experiments: conditional and direct flow, OT, Phikon conditioning, and structural and staining losses.
Andrey Galichin	HER2match implementation and experiments: Direct CFM, LPIPS, stain-aware sampling, staining loss, and error analysis.
Margarita Urazmetova	VP-CFM, crop-size and Exact OT-CFM experiments; MIST data and error audits; DAB- and alignment-aware sampling ablations; quantitative and qualitative evaluation.

Table 4: Team members and their primary contributions.

References

[1] Y. Lipman et al., "Flow Matching for Generative Modeling," ICLR, 2023. https://arxiv.org/abs/2210.02747

[2] A. Tong et al., "Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport," TMLR, 2024. https://arxiv.org/abs/2302.00482

[3] F. Li et al., "Adaptive Supervised PatchNCE Loss for Learning H&E-to-IHC Stain Translation with Inconsistent Groundtruth Image Pairs," MICCAI, 2023. https://doi.org/10.1007/978-3-031-43987-2_61

[4] P. Klockner et al., "GANs vs. Diffusion Models for Virtual Staining with the HER2match Dataset," 2025. https://arxiv.org/abs/2506.18484

[5] P. Klockner et al., "HER2match dataset," Zenodo, 2025. https://doi.org/10.5281/zenodo.15797050

[6] A. Tong et al., "TorchCFM: Conditional Flow Matching library." https://github.com/atong01/conditional-flow-matching